Lecture 10 (Chapter 4)
The Expectation of a Random Variable

Definition Let X be a discrete random variable with p.function f.

The *mean*, *expectation* or *expected value* of X is denoted by $E(X)$ and given by

$$E(X) = \sum_{\text{All } x} x f(x), \quad \text{if}$$

at least one of the following sums is finite:

① $\sum_{\text{Positive } x} x f(x)$; ② $\sum_{\text{Negative } x} x f(x)$.

Remarks

(a) If both sums ① and ② are infinite, ③ is not well defined, and $E(X)$ DOES NOT EXIST

(b) If ① $= \infty$ and ② $< \infty$, then $E(X) = \infty$

If ① $< \infty$ and ② $= -\infty$, then $E(X) = -\infty$.

(c) If ① $< \infty$ and ② $< \infty$, then $E(X) < \infty$.

(d) If X is a bounded random variable, EX always exists.

Example 1 $X \begin{pmatrix} 1 & 2 & 3 & \cdots & n & \cdots \\ \frac{1}{1(1+1)} & \frac{1}{2(2+1)} & \frac{1}{3(3+1)} & \cdots & \frac{1}{n(n+1)} & \cdots \end{pmatrix}$

$$EX = \sum_{n=1}^{\infty} \frac{n}{n(n+1)} = \sum_{n=1}^{\infty} \frac{1}{n+1} = \infty .$$

Example 2 $X \begin{pmatrix} -2 & 0 & 1 & 4 \\ 0.1 & 0.4 & 0.3 & 0.2 \end{pmatrix}$

$EX = (-2)\cdot(0.1) + \ldots + 4\cdot 0.2 = 0.9$, which is NOT one of the possible values of X.

IMPORTANT Two (or many) random variables that may be different but have the same distribution will have the same expected value. This is true no matter the nature of the random variable (discrete, continuous or mixed). Therefore, EX depends only on the distribution of X, and not on the "story" behind X.

We will often say "The expectation of a distribution" instead of "The expectation of a r.v.".

Expectation of continuous distributions

Definition 1 Let X be a bounded r.v. with p.d.f. $f(x)$. Then, the expectation of X is

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx .$$

Example 1 X = Time until a device fails.

Suppose $X \sim f(x)$, with

$$f(x) = \begin{cases} 2x, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then $$E(X) = \int_0^1 x(2x) dx = 2\int_0^1 x^2 dx = \frac{2}{3}.$$

General definition of $E(X)$ for continuous r.v.'s

Assume $X \sim f(x)$ and at least one of

$$\int_0^\infty x f(x)\,dx, \quad \int_{-\infty}^0 x f(x)\,dx \quad \text{is finite.}$$

Then $E(X) \overset{\text{definition}}{=\!=} \int_{-\infty}^{\infty} x f(x)\,dx$.

Example (The Cauchy distribution does not have a mean)

$X \sim f(x) = \dfrac{1}{\pi(1+x^2)}$ ← The Cauchy distribution.

$$\int_0^\infty \frac{x}{\pi(1+x^2)}\,dx = \infty \quad \text{and} \quad \int_{-\infty}^0 \frac{x}{\pi(1+x^2)}\,dx = -\infty,$$

therefore $E(X)$ does not exist.

• See Figures 4.3 and 4.4. page 212.

Figure 4.3 The p.d.f. of a Cauchy distribution.

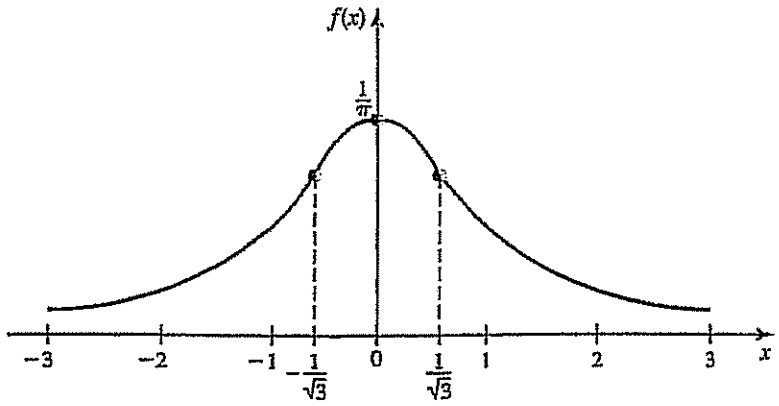
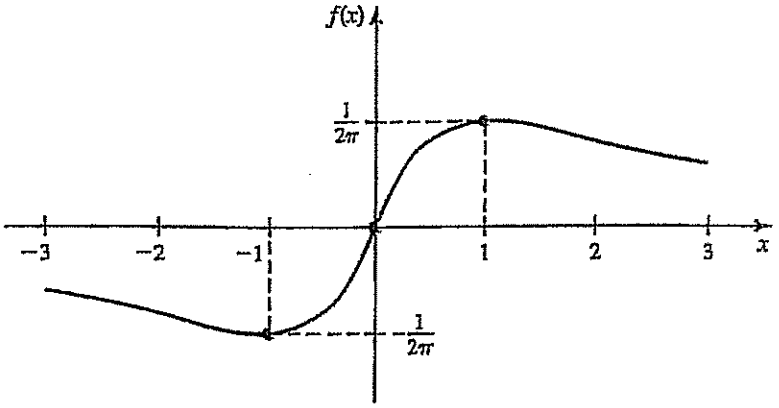


Figure 4.4 The curve $y = xf(x)$ for the Cauchy distribution.



The expectation of a function of a random variable

- Say $X \sim f(x)$ and say that we are interested in $Y = r(X)$, for some given function r. To calculate $E(Y)$ we have two options:

(I) Calculate $g(y)$, the p.f./p.d.f. of the transformed Y, then use the formulas above to calculate EY.

OR

(II) We do not have to actually calculate $g(y)$, in light of the theorem below.

Theorem 4.1.1 Law of the Unconscious Statistician. Let X be a random variable, and let r be a real-valued function of a real variable. If X has a continuous distribution, then

$$E[r(X)] = \int_{-\infty}^{\infty} r(x) f(x)\, dx, \tag{4.1.9}$$

if the mean exists. If X has a discrete distribution, then

$$E[r(X)] = \sum_{\text{All } x} r(x) f(x), \tag{4.1.10}$$

if the mean exists.

Proof A general proof will not be given here. However, we shall provide a proof for two special cases. First, suppose that the distribution of X is discrete. Then the distribution of Y must also be discrete. Let g be the p.f. of Y. For this case,

$$\begin{aligned}\sum_y y g(y) &= \sum_y y \Pr[r(X) = y] \\ &= \sum_y y \sum_{x: r(x)=y} f(x) \\ &= \sum_y \sum_{x: r(x)=y} r(x) f(x) = \sum_x r(x) f(x).\end{aligned}$$

Hence, Eq. (4.1.10) yields the same value as one would obtain from Definition 4.1.1 applied to Y.

Second, suppose that the distribution of X is continuous. Suppose also, as in Sec. 3.8, that $r(x)$ is either strictly increasing or strictly decreasing with differentiable inverse $s(y)$. Then, if we change variables in Eq. (4.1.9) from x to $y = r(x)$,

$$\int_{-\infty}^{\infty} r(x) f(x)\, dx = \int_{-\infty}^{\infty} y f[s(y)] \left| \frac{ds(y)}{dy} \right| dy.$$

It now follows from Eq. (3.8.3) that the right side of this equation is equal to

$$\int_{-\infty}^{\infty} y g(y)\, dy.$$

Hence, Eqs. (4.1.8) and (4.1.9) yield the same value.

Example 2

- X = rate at which iPhones fail / year.
- X is a random variable
- How long will your iPhone last before failure? Look at $Y = \frac{1}{X}$, the time to failure. $E(Y)$ is a good predictor of how long you can still keep your phone.
- Assume $X \sim f(x)$, where

$$f(x) = \begin{cases} 3x^2, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

Once again, for this $f(x)$ say that we are interested in the <u>expected time to failure</u>, that is we want $E\left(\frac{1}{X}\right)$.

<u>Solution 1</u>. Use first the methods in Sect. 3.8 to derive that, for $r(x)=\frac{1}{x}$, the p.d.f. of $Y=r(X)$ is $g(y)=\begin{cases}\frac{3}{y^4}, & \text{if } y>1\\ 0, & \text{otherwise.}\end{cases}$

<u>Then</u> $E(Y)=\int_0^\infty y\cdot\frac{3}{y^4}\,dy=\int_1^\infty \frac{3}{y^3}\,dy$

$$=3\int_1^\infty y^{-3}\,dy=3\cdot\left(\frac{y^{-3+1}}{-3+1}\Bigg|_{y=1}^{y=\infty}\right)$$

$$=3\left(0+\frac{1}{2}\cdot(1)^{-2}\right)=\frac{3}{2}$$

Solution 2 (Use the Law of the unconscious statistician)

$$E(Y) = E\left(\frac{1}{X}\right) = \int_0^\infty \frac{1}{x} \cdot 3x^2 \, dx$$

$$= 3\int_0^1 x \, dx = 3 \cdot \frac{x^2}{2}\Big|_0^1 = \frac{3}{2}. \;//$$

Functions of 2 variables: Expectation

Th. 4.1.2 simplified (see page 215 for full text).

$$E(h(X_1, X_2)) = \iint h(x_1, x_2) f(x_1, x_2) \, dx_1 \, dx_2.$$

N.B. This shows that we do NOT need to determine the distribution of $h(X_1, X_2)$ if we are only interested in $E(h(X_1, X_2))$.

Example 4.1.16 Determining the Expectation of a Function of Two Variables. Suppose that a point (X, Y) is chosen at random from the square S containing all points (x, y) such that $0 \le x \le 1$ and $0 \le y \le 1$. We shall determine the expected value of $X^2 + Y^2$.

Since X and Y have the uniform distribution over the square S, and since the area of S is 1, the joint p.d.f. of X and Y is

$$f(x, y) = \begin{cases} 1 & \text{for } (x, y) \in S, \\ 0 & \text{otherwise.} \end{cases}$$

Therefore,

$$E(X^2 + Y^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x^2 + y^2) f(x, y)\, dx\, dy$$

$$= \int_0^1 \int_0^1 (x^2 + y^2)\, dx\, dy = \frac{2}{3}.$$